

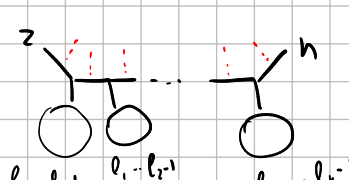
proof that syzygies satisfy all 1-zeros.

• note: syzygy is cyclically invariant $\Rightarrow$ if it satisfies 1 of the 1-zeros it satisfies all of them.

$\rightarrow$ it's enough to show that it vanishes when
$$\begin{cases} X_{2j} = X_{1j} - X_{13} & j=4, \dots, n-1 \\ X_{2n} = -X_{13} \end{cases}$$

D-subsets w.r.t this constraint

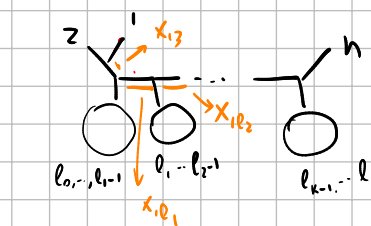
if we have k boundary prop., the D-subset looks like this:



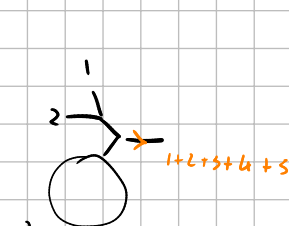
$l_0 = 3 \quad l_k = n$
 $3 = l_0 < l_1 < l_2 < \dots < l_{k-1} < l_k = n$
extending

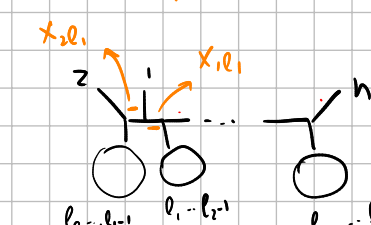
enacting 1 in all possible ways give us the D-subset

• goal: show that D-subset vanishes on constraint surface

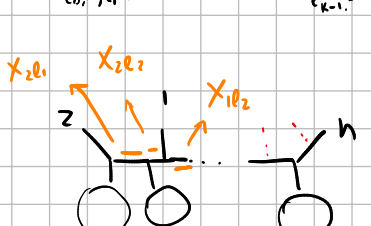


$$= \frac{1}{X_{1l_0} X_{1l_1} X_{1l_2} \dots X_{1l_{k-1}}}$$

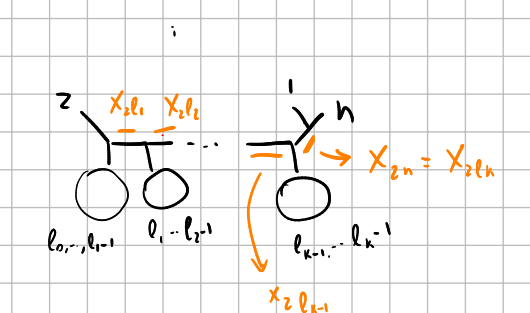




$$= \frac{1}{X_{2l_1} X_{1l_1} X_{1l_2} \dots X_{1l_{k-1}}}$$



$$= \frac{1}{X_{2l_1} X_{2l_2} X_{1l_2} \dots X_{1l_{k-1}}}$$



$$= \frac{1}{X_{2l_1} X_{2l_2} \dots X_{2l_{k-1}} X_{2l_n}}$$

whole D-subst: $\sum_{i=0}^k \frac{1}{\prod_{a=1}^i X_{2le_a}} \frac{1}{\prod_{b=i}^{k-1} X_{1le_b}} = \sum_{i=0}^k \frac{1}{X_{2le_1} \dots X_{2le_i} | X_{1le_i} \dots X_{1le_{k-1}}}$

of X_2

i terms $k-i$ terms

numerator: $\prod_{a=1}^k X_{2le_a} \prod_{b=0}^{k-1} X_{1le_b} = X_{1le_0} \dots X_{1le_{k-1}} X_{2le_1} \dots X_{2le_k}$

D-subst $\propto \sum_{i=0}^k \prod_{a=i+1}^k X_{2le_a} \prod_{b=0}^{i-1} X_{1le_b}$

impose constraints: $\begin{cases} X_{2j} = X_{1j} - X_{13} & j = 4, \dots, n-1 \\ X_{2n} = -X_{13} \end{cases} \rightarrow X_{2le_k} = -X_{1le_0}$

X_{1le_0} appears if $i \geq 1$, X_{2le_k} appears for $i \leq k-1$

$$\begin{aligned} \rightarrow &= \prod_{a=1}^k X_{2le_a} + \prod_{b=0}^{k-1} X_{1le_b} + \sum_{i=1}^{k-1} \prod_{a=i+1}^k X_{2le_a} \prod_{b=0}^{i-1} X_{1le_b} \\ &= X_{2n} \prod_{a=1}^{k-1} X_{2le_a} + X_{13} \prod_{b=1}^{k-1} X_{1le_b} + X_{13} X_{2n} \sum_{i=1}^{k-1} \left(\prod_{a=i+1}^{k-1} X_{2le_a} \right) \left(\prod_{b=1}^{i-1} X_{1le_b} \right) \end{aligned}$$

impose constraints:

$$= -X_{13} \prod_{a=1}^{k-1} (X_{1le_a} - X_{13}) + X_{13} \prod_{b=1}^{k-1} X_{1le_b} - X_{13}^2 \sum_{i=1}^{k-1} \left(\prod_{a=i+1}^{k-1} (X_{1le_a} - X_{13}) \right) \left(\prod_{b=1}^{i-1} X_{1le_b} \right)$$

(partially) Factor out X_{1le_1} :

appears in 1st and 2nd term, and in sum when $i \geq 2$

$$\begin{aligned} &= -X_{13} X_{1le_1} \prod_{a=2}^{k-1} (X_{1le_a} - X_{13}) + X_{13}^2 \prod_{a=2}^{k-1} (X_{1le_a} - X_{13}) + X_{13} X_{1le_1} \prod_{a=2}^{k-1} X_{1le_a} \\ &\quad - X_{13}^2 \prod_{a=2}^{k-1} (X_{1le_a} - X_{13}) + X_{13}^2 X_{1le_1} \sum_{i=2}^{k-1} \left(\prod_{a=i+1}^{k-1} (X_{1le_a} - X_{13}) \right) \left(\prod_{b=2}^{i-1} X_{1le_b} \right) \end{aligned}$$

factor out X_{1le_1}

$$\propto -X_{13} \prod_{a=2}^{k-1} (X_{1le_a} - X_{13}) + X_{13} \prod_{a=2}^{k-1} X_{1le_a} - X_{13}^2 \sum_{i=2}^{k-1} \left(\prod_{a=i+1}^{k-1} (X_{1le_a} - X_{13}) \right) \left(\prod_{b=2}^{i-1} X_{1le_b} \right) \quad (*)$$

all the steps for $(*)$

$$-X_{13} \prod_{a=1}^{k-1} (X_{1\ell_a} - X_{13}) = -X_{13} (X_{1\ell_1} - X_{13}) \prod_{a=2}^{k-1} (X_{1\ell_a} - X_{13})$$

back to $(*)$

→ repeat for $X_{1\ell_2}$

$$-X_{13} \prod_{a=2}^{k-1} (X_{1\ell_a} - X_{13}) + X_{13} \prod_{a=2}^{k-1} X_{1\ell_a} - X_{13}^2 \sum_{i=2}^{k-1} \left(\prod_{a=i+1}^{k-1} (X_{1\ell_a} - X_{13}) \right) \left(\prod_{b=2}^{i-1} X_{1\ell_b} \right)$$

$$= -X_{13} X_{1\ell_2} \prod_{a=3}^{k-1} (X_{1\ell_a} - X_{13}) + X_{13}^2 \prod_{a=3}^{k-1} (X_{1\ell_a} - X_{13}) + X_{13} X_{1\ell_2} \prod_{a=3}^{k-1} X_{1\ell_a}$$

$$-X_{13}^2 \prod_{a=3}^{k-1} (X_{1\ell_a} - X_{13}) - X_{13}^2 X_{1\ell_2} \sum_{i=3}^{k-1} \left(\prod_{a=i+1}^{k-1} (X_{1\ell_a} - X_{13}) \right) \left(\prod_{b=3}^{i-1} X_{1\ell_b} \right)$$

$$\propto -X_{13} \prod_{a=3}^{k-1} (X_{1\ell_a} - X_{13}) + X_{13} \prod_{a=3}^{k-1} X_{1\ell_a} - X_{13}^2 \sum_{i=3}^{k-1} \left(\prod_{a=i+1}^{k-1} (X_{1\ell_a} - X_{13}) \right) \left(\prod_{b=3}^{i-1} X_{1\ell_b} \right)$$

→ every step we increase this index

eventually:

$$\propto -X_{13} \prod_{a=k-1}^{k-1} (X_{1\ell_a} - X_{13}) + X_{13} \prod_{a=k-1}^{k-1} X_{1\ell_a} - X_{13}^2 \sum_{i=k-1}^{k-1} \left(\prod_{a=i+1}^{k-1} (X_{1\ell_a} - X_{13}) \right) \left(\prod_{b=k-1}^{i-1} X_{1\ell_b} \right)$$

$$= -X_{13} (X_{1\ell_{k-1}} - X_{13}) + X_{13} X_{1\ell_{k-1}} - X_{13}^2 = 0$$

$$\prod_{i=1}^3 2 = 2^3$$

$$2 \prod_{i=2}^3 2 = 2 \cdot 2^2$$

$$2 \cdot 2 \cdot \prod_{i=3}^3 2 = 2 \cdot 2 \cdot 2 \cdot \prod_{i=4}^3 2$$

problem for 2-zeros

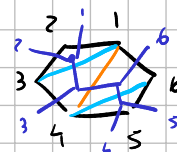
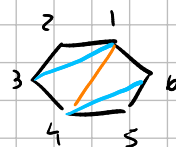
don't know the equivalent of



ex: $n=6$

BP: $X_{14} X_{15} X_{35} X_{36} X_{24} X_{25} X_{26}$

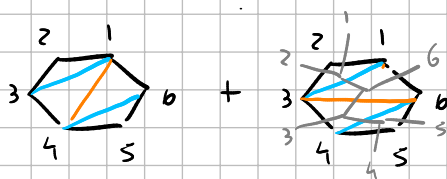
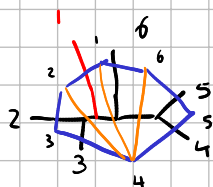
example of graph with 1 boundary prop.



$$\begin{array}{c} 1 \\ | \\ 2 \text{---} 3 \text{---} 4 \text{---} 5 \\ | \\ 6 \end{array} = \frac{1}{X_{13} X_{14} X_{46}} \quad \text{1 BP}$$

what happens if we remove leg 1 and put it back in all ways?

$$\begin{array}{c} 1 \\ | \\ 2 \text{---} 3 \text{---} 4 \text{---} 5 \\ | \\ 6 \end{array} = \frac{1}{X_{24} X_{14} X_{46}} \quad \text{2 BP}$$

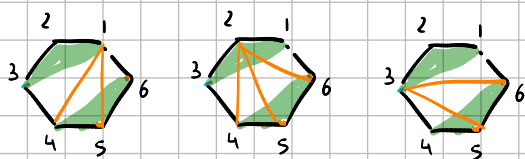


$$\begin{array}{c} 1 \\ | \\ 2 \text{---} 3 \text{---} 4 \text{---} 5 \\ | \\ 6 \end{array} + \begin{array}{c} 1 \\ | \\ 2 \text{---} 3 \text{---} 4 \text{---} 5 \\ | \\ 6 \end{array} = \frac{1}{X_{13} X_{14} X_{46}} + \frac{1}{X_{13} X_{36} X_{46}}$$

$$\begin{array}{c} 1 \\ | \\ 2 \text{---} 3 \text{---} 4 \text{---} 5 \\ | \\ 6 \end{array} \rightarrow \begin{array}{c} 1 \\ | \\ 2 \text{---} 3 \text{---} 4 \text{---} 5 \\ | \\ 6 \end{array} \rightarrow \begin{array}{c} 1 \\ | \\ 2 \text{---} 3 \text{---} 4 \text{---} 5 \\ | \\ 6 \end{array}$$

$$X_{13} X_{14} X_{15} \quad X_{13} X_{35} X_{15} \quad X_{13} X_{36} X_{35}$$

$$X_{14} X_{15} \quad X_{35} X_{36}$$



1 BP not here $13, 46 \rightarrow \frac{1}{X_{14}} + \frac{1}{X_{36}}$

$$1j \rightarrow 3j+1$$

2 BP ① pick 13 as Bern free $\rightarrow 14, 15 \quad 35, 15 \quad 35, 36$

② pick 46 as both free $\rightarrow 14, 24 \quad 24, 26 \quad 26, 36$

⑭ 14 24 15

⑮ 15 25 35

⑳ 24 25 26

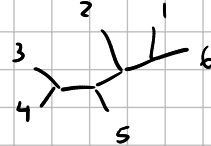
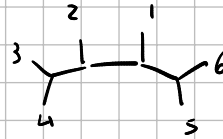
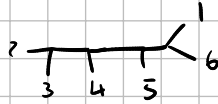
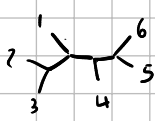
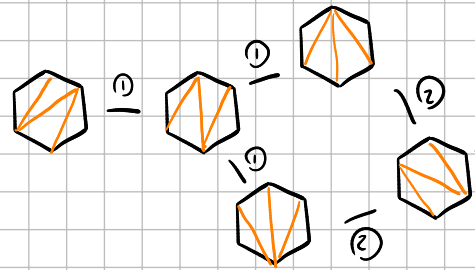
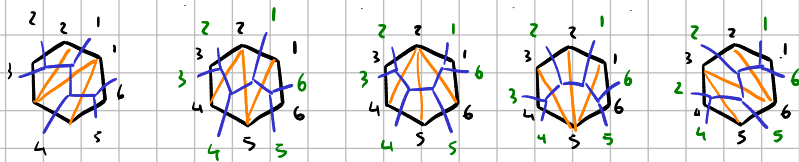


14 15 24 15 25 35 15 25 24

- 24 25 26

- 26 36 35

(26) 26 36 35



$$\frac{1}{X_{14} X_{15}} + \frac{1}{X_{35} X_{15}} + \frac{1}{X_{35} X_{36}} \propto X_{35} X_{36} + X_{14} X_{36} + X_{14} X_{15}$$

insert $X_{36} = -X_{14}$ $X_{35} = X_{15} - X_{14}$

$$\propto (X_{15} - X_{14})(-X_{14}) - X_{14}^2 + X_{14} X_{15} = 0 \quad \checkmark$$